



FLORIDA STATE UNIVERSITY
DEPARTMENT OF SCIENTIFIC COMPUTING

WRITTEN PRELIMINARY EXAMINATION
SUMMER 2024

Question - 7

Submitted By:
Mani Tyagi

Student ID :
200973108
mt22t

May 23, 2024

Contents

1	Maximum Likelihood Estimation	[Dr. Chipilski]	2
1.1	Part (a)		2
1.2	Part (b)		2
1.3	Part (c)		3
1.4	Part (d)		4
1.5	Part (e)		4

1 Maximum Likelihood Estimation[Dr. Chipilski]

1.1 Part (a)

The given probability density function (pdf) is:

$$p(x_i; \mu, \sigma^2) = \frac{1}{x_i \sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right)$$

It is given that the samples are *iid*(independent and identically distributed), so we can express the joint pdf or the Likelihood function as the product of individual pdfs of each of these observations. If there are n observations, the joint pdf can be written as:

$$p(x_1, \dots, x_n; \mu, \sigma^2) = \prod_{i=1}^n p(x_i; \mu, \sigma^2)$$

Substituting the expression for $p(x_i; \mu, \sigma^2)$, we get:

$$p(x_1, \dots, x_n; \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{x_i \sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right)$$

This is our likelihood function.

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{x_i \sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right) \quad (1)$$

1.2 Part (b)

Taking the natural logarithm of the joint pdf in eq.1, we get the log-likelihood function $\ell(\theta; x_1, \dots, x_n)$:

$$\begin{aligned} \ell(\theta; x_1, \dots, x_n) &= \ln L(\theta; x_1, \dots, x_n) \\ &= \ln \left[\prod_{i=1}^n \frac{1}{x_i \sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right) \right] \\ &= \sum_{i=1}^n \ln \left[\frac{1}{x_i \sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right) \right] \quad (\text{using } \ln(ABC\dots) = \ln A + \ln B + \ln C + \dots) \\ &= \sum_{i=1}^n \left[\ln \frac{1}{x_i \sqrt{2\pi\sigma^2}} + \ln \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right) \right] \\ &= \sum_{i=1}^n \left[-\ln x_i \sqrt{2\pi\sigma^2} + \ln \exp\left(-\frac{(\ln x_i - \mu)^2}{2\sigma^2}\right) \right] \\ &= \sum_{i=1}^n \left[-\ln(x_i) - \frac{1}{2} \ln(2\pi\sigma^2) - \frac{(\ln x_i - \mu)^2}{2\sigma^2} \right] \\ &= \sum_{i=1}^n \left[-\ln(x_i) - \frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{(\ln x_i - \mu)^2}{2\sigma^2} \right] \\ &= -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n [\ln(x_i) - \mu]^2 - \sum_{i=1}^n \ln(x_i) \end{aligned}$$

Here, I have used the properties of log and the fact that when a constant term is summed n times, the result can be obtained by multiplying it by n . Hence, we have:

$$\ell(\theta; x_1, \dots, x_n) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n [\ln(x_i) - \mu]^2 - \sum_{i=1}^n \ln(x_i)$$

1.3 Part (c)

The final expression in last part represents the log-likelihood function for the given data and parameters. To find the maximum likelihood estimates (MLEs) of μ and σ^2 , we can differentiate this expression with respect to μ and σ^2 , set the derivatives equal to zero, and solve for μ and σ^2 .

To find $\hat{\mu}$, differentiating l with respect to μ , we get:

$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n [\ln(x_i) - \mu]$$

Setting this equal to zero:

$$\begin{aligned} \frac{1}{\sigma^2} \sum_{i=1}^n [\ln(x_i) - \hat{\mu}] &= 0 \\ \implies \sum_{i=1}^n [\ln(x_i)] - n\hat{\mu} &= 0 \end{aligned}$$

$$\implies \hat{\mu} = \frac{1}{n} \sum_{i=1}^n \ln(x_i)$$

To find $\hat{\sigma}^2$, we can differentiate l with respect to σ^2 :

$$\frac{\partial l}{\partial (\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n [\ln(x_i) - \mu]^2$$

Setting this equal to zero:

$$-\frac{n}{2\hat{\sigma}^2} + \frac{1}{2(\hat{\sigma}^2)^2} \sum_{i=1}^n [\ln(x_i) - \hat{\mu}]^2 = 0$$

$$\text{On rearranging, we have: } n = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n [\ln(x_i) - \hat{\mu}]^2 = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n [\ln(x_i) - \hat{\mu}]^2$$

Finally, we obtain the MLE expressions for $\hat{\sigma}^2$:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n [\ln(x_i) - \hat{\mu}]^2 = \frac{1}{n} \sum_{i=1}^n \left[\ln(x_i) - \frac{1}{n} \sum_{i=1}^n \ln(x_i) \right]^2$$

Complete MLE expression:

$$\hat{\theta} = [\hat{\mu} \quad \hat{\sigma}^2]^T = \left[\frac{1}{n} \sum_{i=1}^n \ln(x_i) \quad \frac{1}{n} \sum_{i=1}^n \left(\ln(x_i) - \frac{1}{n} \sum_{i=1}^n \ln(x_i) \right)^2 \right]^T$$

1.4 Part (d)

The obtained results are

MLE estimate of Mean 2.95474

MLE estimate of Variance: 0.97423

Hence, $\hat{\theta} = [\hat{\mu} \ \hat{\sigma}^2]^T = [2.95474 \ 0.97423]^T$

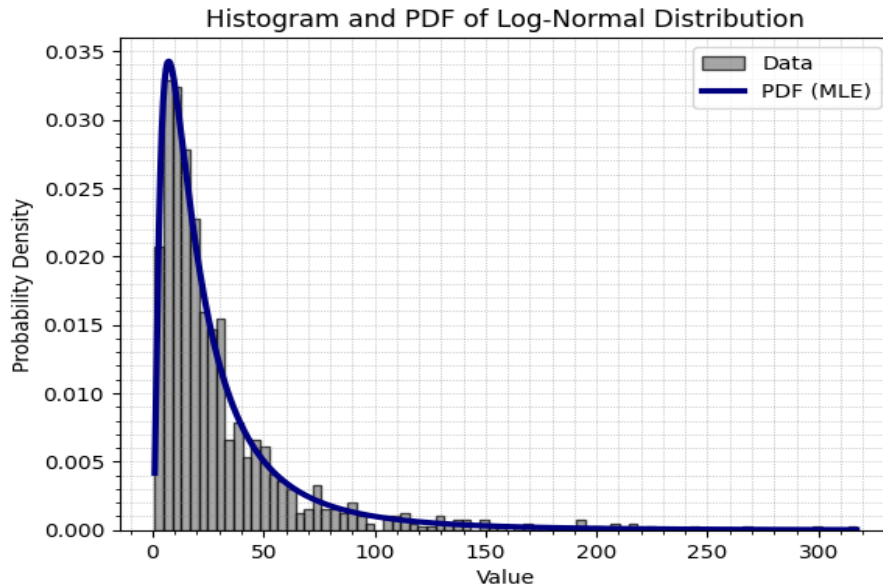


Figure 1: Histogram of *lognormal_sample.txt* with MLE pdf overlaid.

The MLE pdf is displaying the same behaviour to that of data. This shows that the calculated MLE expression obtained is correct and a good fit for the data.

1.5 Part (e)

The obtained results using gradient descent optimization used are:

RESULTS

Iterations: 89

mu: 2.95474

sigma: 0.97424

Hence, $\hat{\theta} = [\hat{\mu} \ \hat{\sigma}^2]^T = [2.95474 \ 0.97424]^T$

It takes 89 iterations to converge to the analytical MLE estimates calculated in part (d). As shown in Fig-2, $\tilde{l}$ and μ converges smoothly from their initial guesses. However, σ^2 first deviates and starts increasing and then decreases to converged value. This might be because of sub optimal initial guess

of direction. Also, the convergence depend greatly on initial values and step-size α . For good initial guesses (like in this case), the solution will converge in lesser number of iterations.

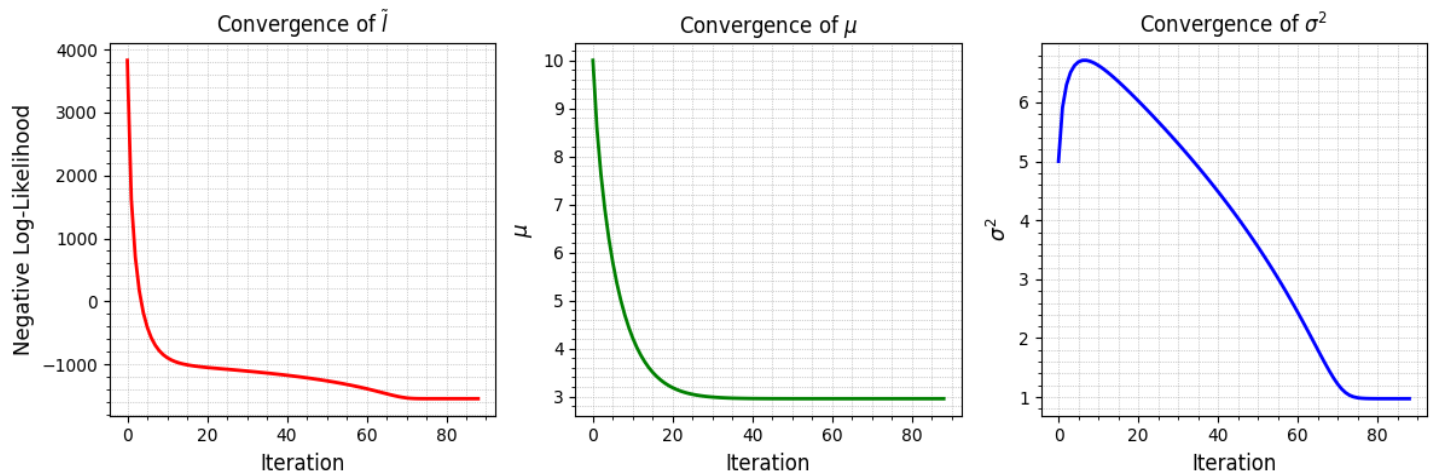


Figure 2: Numerical results of MLE

References

- ACSII Lecture Notes, Dr. Plewa
- Data Assimilation Lecture Notes, Dr. Chipilski
- Python documentation: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.lognorm.html>
- Gradient descent algorithm: https://en.wikipedia.org/wiki/Gradient_descent